Work in progress — draft, not peer reviewed

A Lean-Verified Conditional Uniform Ceiling for a Cubic-Tethered Vorticity Model

Conditional results; draft — public repository, not peer reviewed

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Scope. This paper concerns a *modified* vorticity equation with an added cubic damping tether. That model is **not** Navier–Stokes. The main result is *conditional* on the named Prop `FieldStretchingLogLoss`, which is **not proved**, and on six further hypotheses. The paper does **not** solve the Clay Millennium Problem, and it says nothing about regularity of unmodified 3D Navier–Stokes.

Abstract

In the Lean 4/Mathlib [2, 4] repository `metriplectic-world-model` (commit `2274a23`), we prove a uniform-in-time ceiling on $\sup|\omega|$ for a vorticity model with cubic tether $-\kappa|\omega|^2\omega$ (`field_uniform_ceiling_logLoss_cubicTether`). It is conditional on a pointwise log-loss stretching Prop and on six regularity, support and sign hypotheses, which we list verbatim. We explain why the known Calderón–Zygmund estimate does not supply the stretching Prop. The axioms used are only `propext`, `Classical.choice` and `Quot.sound`.

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Keywords. vorticity equation; cubic damping; maximum principle; Danskin's theorem; Lean 4; formal verification; work in progress.

1 Introduction

Adding a cubic damping term to the vorticity equation produces a model in which the peak vorticity $Y(t) = \sup_x|\omega(t,x)|$ satisfies a comparison inequality of the form $Y' \le CY^2 - \kappa Y^3$, which yields a uniform-in-time ceiling. The difficulty lies in the vortex-stretching term, whose known bounds involve norms stronger than Y. In this paper we isolate exactly what is needed: we prove, in Lean 4, a uniform ceiling for the cubic-tethered model conditional on a Y-only logarithmic stretching hypothesis and six explicit regularity and support hypotheses, and we explain why the classical Calderón–Zygmund estimate does not supply the stretching hypothesis.

2 The cubic-tethered vorticity model

The model is `ModelVorticityPDE` $\nu\ \omega\ u\ T$:

$$u = \mathrm{BS}(\omega), \qquad \operatorname{div}\omega = 0, \qquad \partial_t\omega + (u\cdot\nabla)\omega - (\omega\cdot\nabla)u = \nu\Delta\omega + T. \tag{2.1}$$

It uses the cubic tether $T =$ `cubicTether` $\kappa\ \omega$, i.e. $T = -\kappa|\omega|^2\omega$. This is *not* Navier–Stokes; the tether is an added damping term. Write

$$Y(t) = \sup_x|\omega(t,x)|, \qquad \mathrm{ceiling}(M_0, C, \kappa) = \max\bigl(M_0, \max(C/\kappa, 0)\bigr).$$

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Named hypothesis 2.1 (`FieldStretchingLogLoss` ω u Y C_1 C_2[†]; **not proved**). For every $t \geq 0$ and every global maximizer x_* of $|\omega(t,\cdot)|$:

$$\omega \cdot (\omega \cdot \nabla) u\,(t, x_*) \leq C_1 Y^3 + C_2 Y^3 \log(1+Y). \tag{2.2}$$

Theorem 2.2 (`field_uniform_ceiling_logLoss_cubicTether`[†]; conditional). *Assume all of the following:*

1. $\nu \geq 0$, $\kappa > 0$, $C_2 \geq 0$;
2. `ModelVorticityPDE` ν ω u (`cubicTether` κ ω);
3. *a* fixed *compact nonempty set* K *with* $\omega(s,x) = 0$ *for* $x \notin K$, *for all* s;
4. $\omega(t) \in C^2$ *in space for every* t;
5. *joint continuity of* ω *and of* $\partial_t \omega$, *and the pointwise time derivative* $\partial_t \omega(s,x)$ *exists for all* s, x;
6. $Y(s)$ *is the least upper bound of* $\{|\omega(s,x)|\}$ *for all* s;
7. `FieldStretchingLogLoss` ω u Y C_1 C_2.

Then for all $t \geq 0$,

$$Y(t) \leq \mathrm{ceiling}\Big(Y(0),\ C_1 + \frac{\kappa}{4} + \frac{4C_2^2}{\kappa},\ \frac{3\kappa}{4}\Big). \tag{2.3}$$

Proof route.

- `fieldNetRate_of_logLoss_cubicTether`[†] uses $\log(1+y) \leq 2\sqrt{1+y}$ and Young's inequality to absorb the log loss into the quartic tether. This gives `FieldNetRate` with $C = C_1 + \kappa/4 + 4C_2^2/\kappa$ and $\kappa_{\mathrm{eff}} = 3\kappa/4$.
- `field_uniform_ceiling_of_netRate`[†] applies a Danskin upper-Dini bound for Y^2 at a maximizer. There the transport term vanishes and $\omega \cdot \Delta\omega \leq 0$.
- A Dini comparison lemma, `ceiling_of_upper_dini`[†], finishes the proof. No differentiability of Y is assumed.

Why the known Calderón–Zygmund estimate does not suffice. The classical log estimate (Beale–Kato–Majda [1], Kozono–Taniuchi [3] type) reads

$$\|\nabla u\|_{L^\infty} \leq C\Big(1 + Y\big(1 + \log^+(\|\omega\|_{C^\alpha}/Y)\big) + \|\omega\|_{L^2}\Big). \tag{2.4}$$

The logarithm contains a *higher* norm, $\|\omega\|_{C^\alpha}$. The ceiling argument controls only Y, and a time-uniform bound on $\|\omega(t)\|_{C^\alpha}$ is essentially the regularity question itself. So the known estimate does not imply `FieldStretchingLogLoss`, whose Y-only form is a strictly stronger hypothesis than anything known.

The CZ estimate itself is also not formalized in that repository, which has no singular-integral or Hölder theory for ∇BS. The existence of solutions with uniformly fixed compact support (hypothesis 3) is likewise not established. Neither the theorem nor anything else in this paper says anything about unmodified Navier–Stokes.

Lean hypotheses, verbatim. The binders of `field_uniform_ceiling_logLoss_cubicTether` are:

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```
(hν : 0 ≤ ν) (hκ : 0 < κ) (hC₂ : 0 ≤ C₂)
(hpde : ModelVorticityPDE ν ω u (cubicTether κ ω))
{K : Set T3} (hK : IsCompact K) (hne : K.Nonempty)
(hsupp : ∀ s x, x ∉ K → ω s x = 0)
(hreg : ∀ t, ContDiff ℝ 2 (ω t))
(hc : Continuous (Function.uncurry ω))
(hct : Continuous (Function.uncurry (time_deriv ω)))
(hd : ∀ s x, HasDerivAt (fun r => ω r x) (time_deriv ω s x) s)
(hY : ∀ s, IsLUB (range fun x => ‖ω s x‖) (Y s))
(hS : FieldStretchingLogLoss ω u Y C₁ C₂)
```

3 Stated vs. proved

Statement	Status
Log loss + cubic tether ⇒ `FieldNetRate`, `fieldNetRate_of_logLoss_cubicTether`	proved unconditionally (an implication between Props)
Dini ceiling, `field_uniform_ceiling_of_netRate`	proved conditionally on `FieldNetRate` and hypotheses 1–6
Uniform ceiling, `field_uniform_ceiling_logLoss_cubicTether`	proved conditionally on `FieldStretchingLogLoss` and hypotheses 1–6
`FieldStretchingLogLoss`	not proved
CZ/BKM log estimate for ∇BS	not proved (not formalized); would not imply `FieldStretchingLogLoss` anyway
Existence of solutions with fixed compact support	not proved
Anything about unmodified NS / Clay	not proved, not claimed

A Theorem-to-Lean map

Names marked † were coined during formalization and are not the author's terminology; they may be renamed later.

Lean name	File	Repo @ commit
`field_uniform_ceiling_logLoss_cubicTether`†	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ `2274a23`
`fieldNetRate_of_logLoss_cubicTether`†	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ `2274a23`
`field_uniform_ceiling_of_netRate`†	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ `2274a23`
`FieldStretchingLogLoss`†, `FieldNetRate`† (defs)	`MetripelecticWorldModel/LogLossCeiling.lean`	mwm @ `2274a23`
`cubicTether`†, `fieldTetherFeedback_cubicTether`†	`MetripelecticWorldModel/DiniCeiling.lean`	mwm @ `2274a23`
`ceiling` (def)	`MetripelecticWorldModel/Ceiling.lean`	mwm @ `2274a23`
`ModelVorticityPDE`† (def)	`MetripelecticWorldModel/SmoothOrbit.lean`	mwm @ `2274a23`

mwm = `metriplectic-world-model`.

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B Axiom audit

At commit `2274a23`, `lake build` succeeds with 0 `sorry`. `lake env lean AxiomAudit.lean` (765 theorems; see the counting convention below) throws an error if any theorem uses an axiom other than `propext`, `Classical.choice` or `Quot.sound`, and it completes without error. Every theorem above is in the audit.

Counting convention. The figure 765 is the *kernel count*: the number of non-internal theorem constants in the module trees `MetriplecticWorldModel`, `FrohmanianSymplecticTether` and `TetherMayForm`, as printed by `AxiomAudit.lean` (`TOTAL theorems=765 nonstandard=0`). For comparison, the *source count* of declarations introduced by `theorem` or `lemma` in the tracked `.lean` files, excluding `AxiomAudit.lean`, is 342; the difference consists of theorems generated by Lean and by declaration commands other than `theorem`/`lemma`.

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References

[1] J. T. Beale, T. Kato, and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys. **94** (1984), no. 1, 61–66.

[2] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Comput. Sci., vol. 12699, Springer, 2021, pp. 625–635.

[3] H. Kozono and Y. Taniuchi, *Limiting case of the Sobolev inequality in BMO, with application to the Euler equations*, Comm. Math. Phys. **214** (2000), no. 1, 191–200.

[4] The mathlib Community, *The Lean mathematical library*, Proc. 9th ACM SIGPLAN Int. Conf. Certified Programs and Proofs (CPP 2020), 367–381, 2020.